

Virus - microbiota interaction: Antibiotic-induced gut dysbiosis exacerbates fowl adenovirus replication and systemic pathology in broiler chickens

Miguel Matos¹, Fatou Geissler¹, Barbara Jaskulska¹, Amalia Bogri², Antton Alberdi Estibaritz², Michael Hess¹

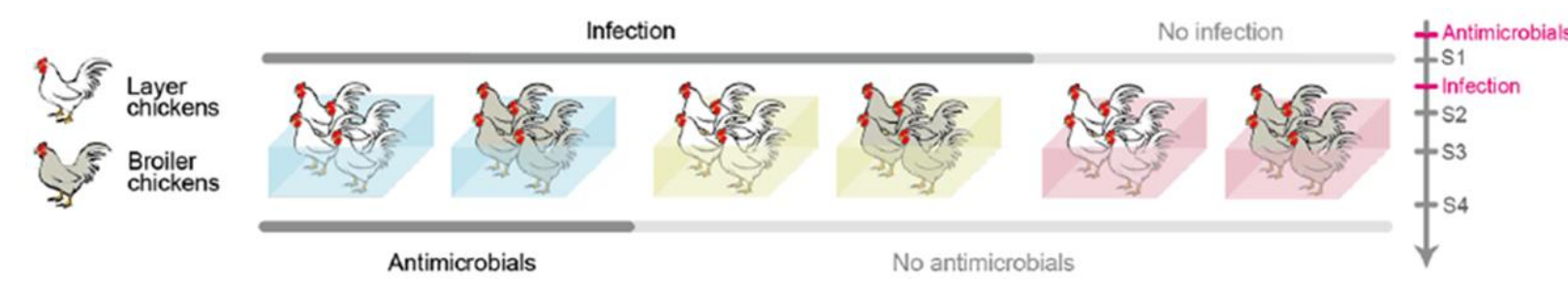
¹ Clinical Unit for Poultry Medicine, University of Veterinary Medicine Vienna, Vienna, Austria

² Center for Evolutionary Hologenomics, Globe Institute, University of Copenhagen, Copenhagen, Denmark

Introduction

Fowl adenoviruses (FAdVs) cause inclusion body hepatitis (IBH), with broiler chickens consistently more susceptible than layers. While viral and host factors influence disease outcome, the contribution of gut microbiota integrity remains poorly defined. Because FAdVs enter via the faecal–oral route and initially replicate in the intestine, we hypothesized that antibiotic-induced gut dysbiosis facilitates systemic viral spread, enhances replication in target organs, and exacerbates pathology in a host-dependent manner. To test this, we investigated antimicrobial-mediated microbiota disruption and FAdV pathogenesis in broiler and layer chickens.

Experimental design



Group	Bird (n)	Treatment ¹	Oral infection ²	Sampling (dpi)
TF1-B	SPF broilers (25)	no	yes	0, 3, 5, 7, 14
TF2-B	SPF broilers (25)	yes	yes	
TF3-B	SPF broilers (25)	no	no	
TF1-L	SPF layers (23)	no	yes	
TF2-L	SPF layers (23)	yes	yes	
TF3-L	SPF layers (23)	no	no	

¹ enrofloxacin 10 mg/kg/day (water) between 9-14 days of age

² 13/18153 strain (FAdV-8b): 0.5 mL/bird (10⁷ TCID₅₀/mL) at 14 days of age

Study readouts

- Body weight monitoring
- Organ-to-body weight ratios
- Histopathology
- Immunohistochemistry
- Quantitative PCR (viral load)
- Caecal metagenomics

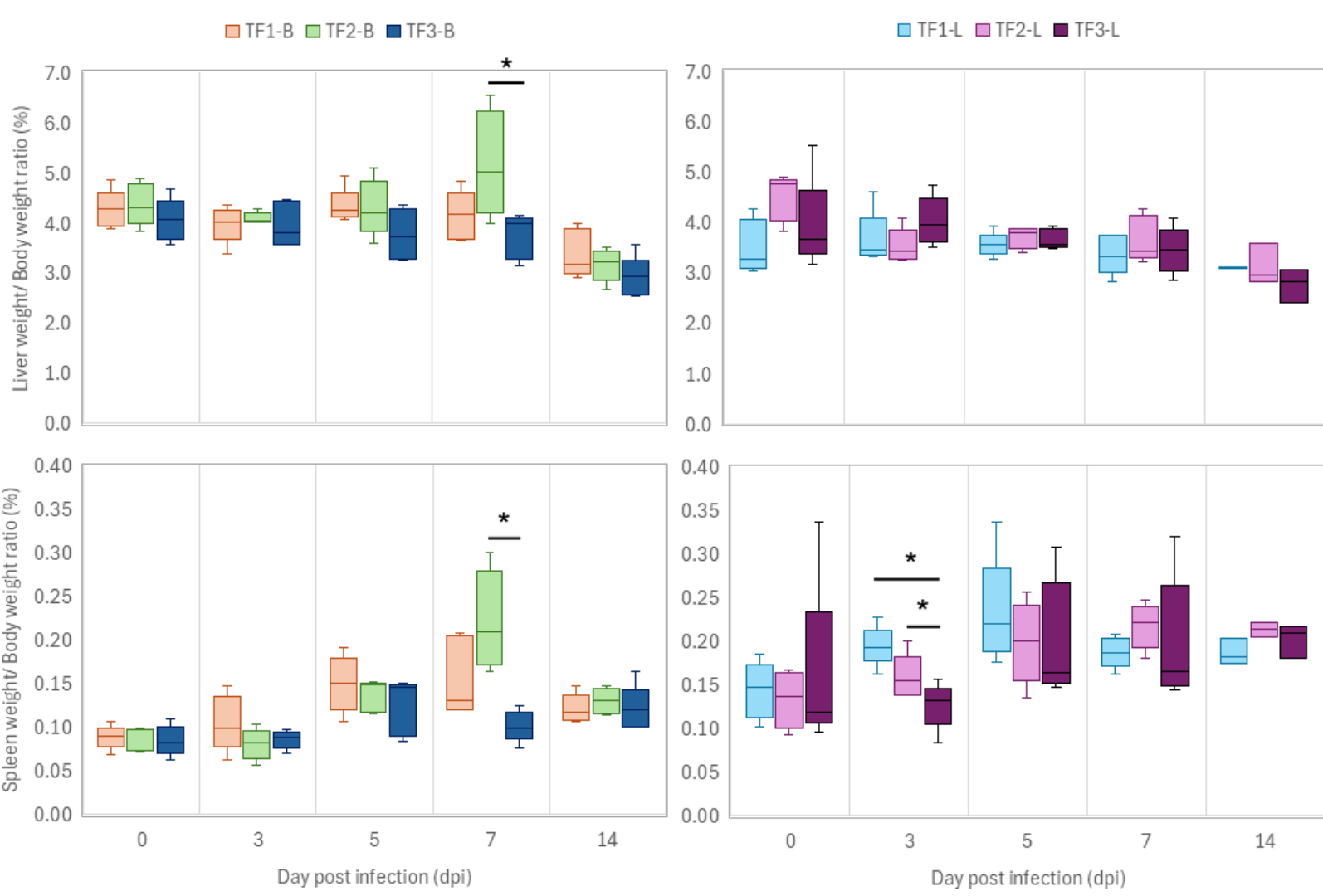
Results

Host response

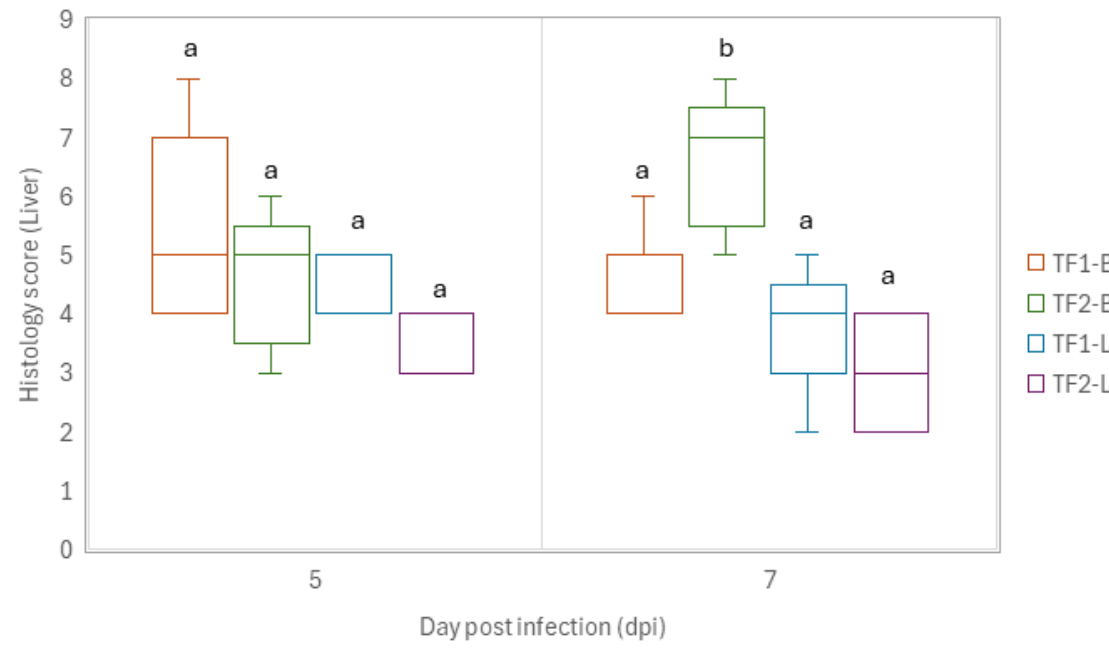
Body weight change. FAdV infection caused a transient reduction in weight gain in broilers, most pronounced at 3–5 dpi after antimicrobial treatment. Layers showed minimal changes, indicating greater susceptibility of broilers to infection-associated performance effects.



Organ-to-body weight ratios. Hepatomegaly and splenomegaly peaked at 7 dpi and were most pronounced in antimicrobial-treated broilers. Layers showed only mild and transient changes, consistent with greater systemic involvement in broilers during FAdV infection.



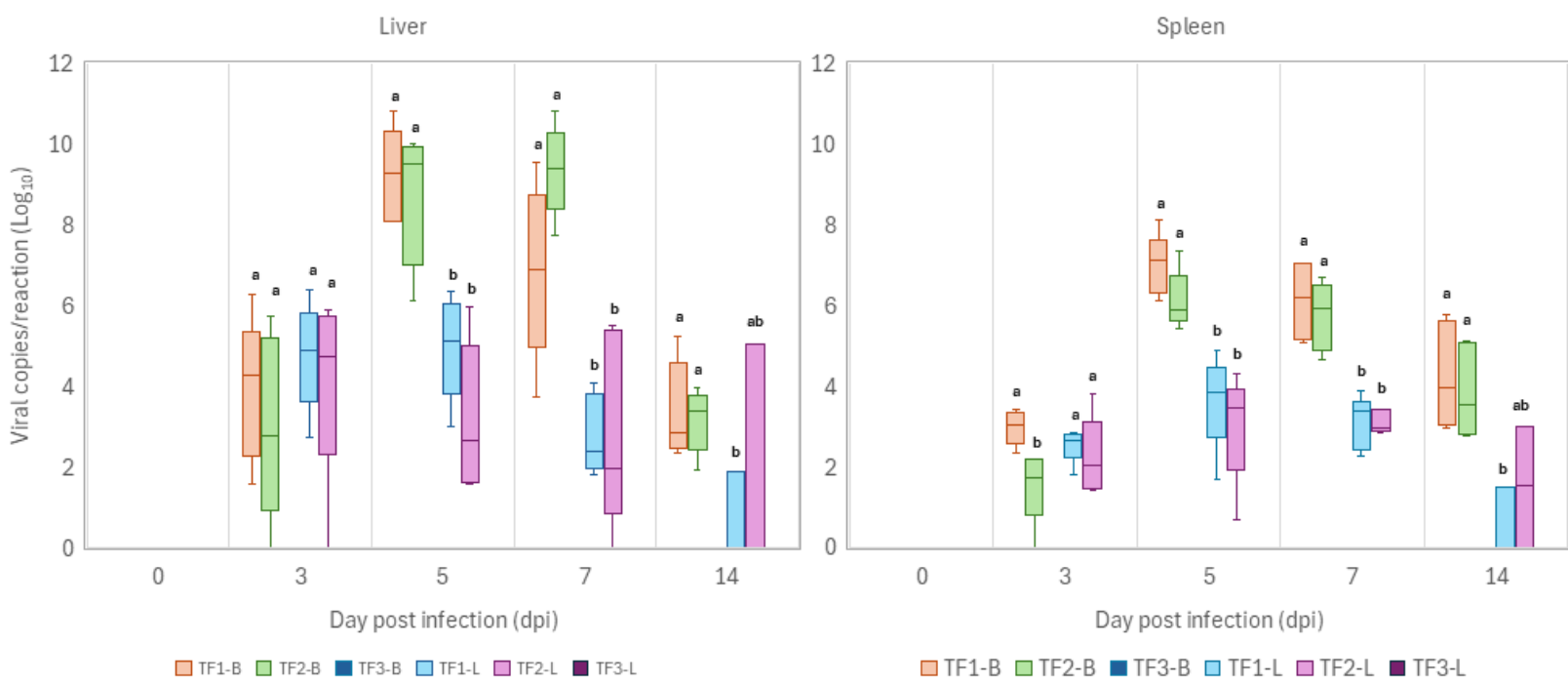
Histopathology & Immunohistochemistry. Hepatic lesion scores peaked at 7 dpi and were highest in antimicrobial-treated broilers. Immunohistochemistry confirmed abundant viral antigen in affected hepatocytes, whereas layers showed only mild and transient lesions.



Viral replication & Microbiome

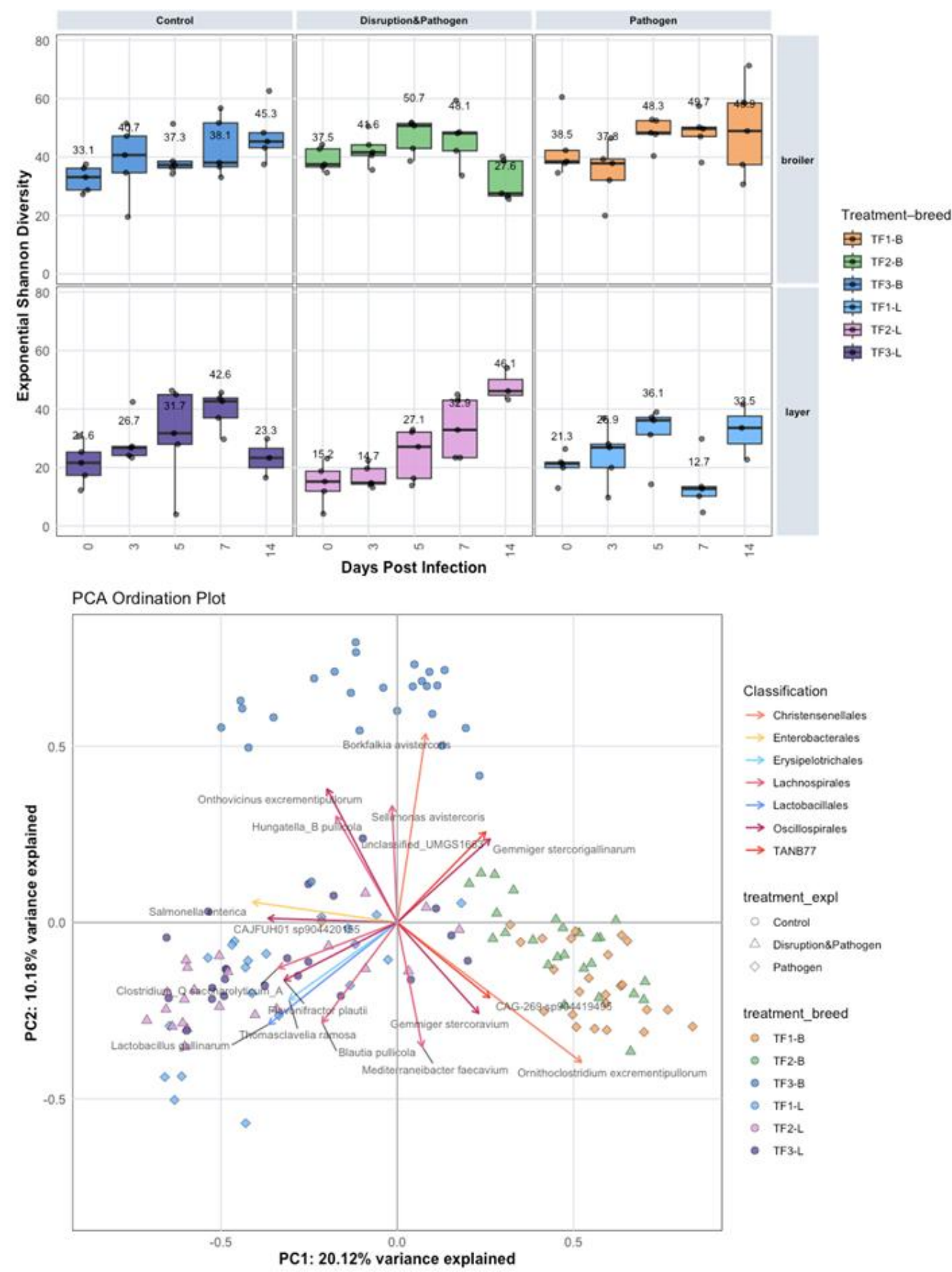
Viral load.

Quantitative PCR demonstrated systemic FAdV replication in liver and spleen. Viral loads peaked at 5–7 dpi and were consistently higher in broilers than in layers. Antimicrobial-treated broilers showed the highest viral loads in liver, indicating that microbiota disruption enhances viral replication.



Metagenomics.

Caecal metagenomic analysis of antimicrobial-treated birds revealed transient spikes in bacterial load and diversity, likely due to community destabilisation and opportunistic blooms following the collapse of commensal populations. Broilers exhibited substantially more bacterial interactions than layers, and antimicrobial-treated samples showed a pronounced collapse in bacterial network connectivity.



Key conclusions

- Antibiotic-induced dysbiosis promotes systemic FAdV replication.
- Broilers show greater susceptibility to infection than layers.
- Higher viral loads coincide with increased organ pathology.
- Microbiota integrity is a key modulator of adenovirus pathogenesis.

These findings highlight the importance of microbiota integrity in fowl adenovirus infections.